# Complete preference frontiers and numerical conventions

These sections are copied verbatim from the preceding Supplement. Every frontier region, boundary tie and numerical convention remains present; the separate PDF is a page-budget relocation, not a new analysis or edited result. The compact Supplement links this file through the extended-results index.

## E1 Complete preference frontier: full sample

Open intervals exclude their endpoints. Point rows retain all ties, including ties at the extreme loss weights. Method codes here denote the original fixed report menu; their definitions remain in the main methods table. The sparse grid is a sensitivity on the same record, not independent data.

Table E1: Complete saved preference frontier, full sample. Exact ties prevent an artificially unique winner.

| Task | Kind | Left / point $a$ | Right $a$ | Minimizers |
|---|---|---|---|---|
| FX full H6/W720 | open | $\frac{0}{1}$ | $\frac{105760946788801400}{110990640317093439}$ | QR |
| | open | $\frac{105760946788801400}{110990640317093439}$ | $\frac{19056123729472220}{19757413338275541}$ | KNN |
| | open | $\frac{19056123729472220}{19757413338275541}$ | $\frac{1}{1}$ | frozen_S |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{105760946788801400}{110990640317093439}$ | | KNN, QR |
| | point | $\frac{19056123729472220}{19757413338275541}$ | | KNN, frozen_S |
| | point | $\frac{1}{1}$ | | frozen_S |
| FX full H6/W2880 | open | $\frac{0}{1}$ | $\frac{34631329313121430}{35599009063461303}$ | QR |
| | open | $\frac{34631329313121430}{35599009063461303}$ | $\frac{1}{1}$ | SCALED_KNN |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{34631329313121430}{35599009063461303}$ | | QR, SCALED_KNN |
| | point | $\frac{1}{1}$ | | SCALED_KNN |
| FX full H24/W720 | open | $\frac{0}{1}$ | $\frac{32241742214205920}{38133893639725449}$ | QR |
| | open | $\frac{32241742214205920}{38133893639725449}$ | $\frac{92050107608238600}{103514765282707447}$ | KNN |
| | open | $\frac{92050107608238600}{103514765282707447}$ | $\frac{29075431634708860}{30985073840047543}$ | AR1_REPLAY |
| | open | $\frac{29075431634708860}{30985073840047543}$ | $\frac{1}{1}$ | frozen_S |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{32241742214205920}{38133893639725449}$ | | KNN, QR |

Table E1: Continued.

| Task | Kind | Left / point $a$ | Right $a$ | Minimizers |
|---|---|---|---|---|
| | point | $\frac{92050107608238600}{103514765282707447}$ | | AR1_REPLAY, KNN |
| | point | $\frac{29075431634708860}{3098507384004754 3}$ | | AR1_REPLAY, frozen_S |
| | point | $\frac{1}{1}$ | | frozen_S |
| FX full H24/W2880 | open | $\frac{0}{1}$ | $\frac{335769696945034900}{350844274558596259}$ | QR |
| | open | $\frac{335769696945034900}{350844274558596259}$ | $\frac{1}{1}$ | frozen_S |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{335769696945034900}{350844274558596259}$ | | QR, frozen_S |
| | point | $\frac{1}{1}$ | | frozen_S |
| Bills full H6/W52 | open | $\frac{0}{1}$ | $\frac{3320065684462664}{4624714213835193}$ | QR |
| | open | $\frac{3320065684462664}{4624714213835193}$ | $\frac{156305902436775700}{160668673434229883}$ | AR1_REPLAY |
| | open | $\frac{156305902436775700}{160668673434229883}$ | $\frac{1}{1}$ | KNN, frozen_F |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{3320065684462664}{4624714213835193}$ | | AR1_REPLAY, QR |
| | point | $\frac{156305902436775700}{160668673434229883}$ | | AR1_REPLAY, KNN, frozen_F |
| | point | $\frac{1}{1}$ | | KNN, frozen_F |
| Bills full H6/W260 | open | $\frac{0}{1}$ | $\frac{8679397304452847}{9039550431967810}$ | AR1_REPLAY |
| | open | $\frac{8679397304452847}{9039550431967810}$ | $\frac{1}{1}$ | frozen_K |
| | point | $\frac{0}{1}$ | | AR1_REPLAY |
| | point | $\frac{8679397304452847}{9039550431967810}$ | | AR1_REPLAY, frozen_K |
| | point | $\frac{1}{1}$ | | frozen_K |
| Bills full H24/W52 | open | $\frac{0}{1}$ | $\frac{5759715075028296}{13586893226278237}$ | QR |
| | open | $\frac{5759715075028296}{13586893226278237}$ | $\frac{117611155234657025}{130987328519855581}$ | frozen_K |
| | open | $\frac{117611155234657025}{130987328519855581}$ | $\frac{1}{1}$ | frozen_S |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{5759715075028296}{13586893226278237}$ | | QR, frozen_K |

Table E1: Continued.

| Task | Kind | Left / point $a$ | Right $a$ | Minimizers |
|---|---|---|---|---|
| | point | $\frac{117611155234657025}{130987328519855581}$ | | frozen_K, frozen_S |
| | point | $\frac{1}{1}$ | | frozen_S |
| Bills full H24/W260 | open | $\frac{0}{1}$ | $\frac{3813577886766790}{4899469998019859}$ | frozen_K |
| | open | $\frac{3813577886766790}{4899469998019859}$ | $\frac{91333672611602885}{94156776268488263}$ | AR1_REPLAY |
| | open | $\frac{91333672611602885}{94156776268488263}$ | $\frac{240827075812274250}{241875451263537787}$ | KNN |
| | open | $\frac{240827075812274250}{241875451263537787}$ | $\frac{1}{1}$ | frozen_S |
| | point | $\frac{0}{1}$ | | frozen_K |
| | point | $\frac{3813577886766790}{4899469998019859}$ | | AR1_REPLAY, frozen_K |
| | point | $\frac{91333672611602885}{94156776268488263}$ | | AR1_REPLAY, KNN |
| | point | $\frac{240827075812274250}{241875451263537787}$ | | KNN, frozen_S |
| | point | $\frac{1}{1}$ | | frozen_F, frozen_S |

## E2 Complete preference frontier: sparse sensitivity

Open intervals exclude their endpoints. Point rows retain all ties, including ties at the extreme loss weights. Method codes here denote the original fixed report menu; their definitions remain in the main methods table. The sparse grid is a sensitivity on the same record, not independent data.

Table E2: Complete saved preference frontier, fixed sparse grid. Exact ties prevent an artificially unique winner.

| Task | Kind | Left / point $a$ | Right $a$ | Minimizers |
|---|---|---|---|---|
| FX sparse H6/W720 | open | $\frac{0}{1}$ | $\frac{35228646474289300}{37058573227618587}$ | QR |
| | open | $\frac{35228646474289300}{37058573227618587}$ | $\frac{128015277909259100}{130046193358911709}$ | KNN |
| | open | $\frac{128015277909259100}{130046193358911709}$ | $\frac{1}{1}$ | SCALED_KNN |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{35228646474289300}{37058573227618587}$ | | KNN, QR |
| | point | $\frac{128015277909259100}{130046193358911709}$ | | KNN, SCALED_KNN |
| | point | $\frac{1}{1}$ | | SCALED_KNN |

Table E2: Continued.

| Task | Kind | Left / point $a$ | Right $a$ | Minimizers |
|---|---|---|---|---|
| FX sparse H6/W2880 | open | $\frac{0}{1}$ | $\frac{59571292631784500}{60902063609752461}$ | QR |
| | open | $\frac{59571292631784500}{60902063609752461}$ | $\frac{1}{1}$ | SCALED_KNN |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{59571292631784500}{60902063609752461}$ | | QR, SCALED_KNN |
| | point | $\frac{1}{1}$ | | SCALED_KNN |
| FX sparse H24/W720 | open | $\frac{0}{1}$ | $\frac{7530906001239450}{9350542730727077}$ | QR |
| | open | $\frac{7530906001239450}{9350542730727077}$ | $\frac{18652943561940400}{20490003118958463}$ | KNN |
| | open | $\frac{18652943561940400}{20490003118958463}$ | $\frac{11935316416561250}{12439911923115111}$ | AR1_REPLAY |
| | open | $\frac{11935316416561250}{12439911923115111}$ | $\frac{1}{1}$ | frozen_S |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{7530906001239450}{9350542730727077}$ | | KNN, QR |
| | point | $\frac{18652943561940400}{20490003118958463}$ | | AR1_REPLAY, KNN |
| | point | $\frac{11935316416561250}{12439911923115111}$ | | AR1_REPLAY, frozen_S |
| | point | $\frac{1}{1}$ | | frozen_S |
| FX sparse H24/W2880 | open | $\frac{0}{1}$ | $\frac{578937865306180}{598942665753459}$ | QR |
| | open | $\frac{578937865306180}{598942665753459}$ | $\frac{1}{1}$ | frozen_S |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{578937865306180}{598942665753459}$ | | QR, frozen_S |
| | point | $\frac{1}{1}$ | | frozen_S |
| Bills sparse H6/W52 | open | $\frac{0}{1}$ | $\frac{1315902266754605}{2037515146643951}$ | QR |
| | open | $\frac{1315902266754605}{2037515146643951}$ | $\frac{679337}{682434}$ | frozen_K |
| | open | $\frac{679337}{682434}$ | $\frac{1}{1}$ | KNN, frozen_F |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{1315902266754605}{2037515146643951}$ | | QR, frozen_K |
| | point | $\frac{679337}{682434}$ | | KNN, frozen_F, frozen_K |

Table E2: Continued.

| Task | Kind | Left / point $a$ | Right $a$ | Minimizers |
|---|---|---|---|---|
| | point | $\frac{1}{1}$ | | KNN, frozen_F |
| Bills sparse H6/W260 | open | $\frac{0}{1}$ | $\frac{39729652848099850}{42395294149611417}$ | AR1_REPLAY |
| | open | $\frac{39729652848099850}{42395294149611417}$ | $\frac{1}{1}$ | frozen_K |
| | point | $\frac{0}{1}$ | | AR1_REPLAY |
| | point | $\frac{39729652848099850}{42395294149611417}$ | | AR1_REPLAY, frozen_K |
| | point | $\frac{1}{1}$ | | frozen_K |
| Bills sparse H24/W52 | open | $\frac{0}{1}$ | $\frac{8631914270543611}{12443891775419019}$ | QR |
| | open | $\frac{8631914270543611}{12443891775419019}$ | $\frac{2529092855837558}{2690148205992001}$ | frozen_K |
| | open | $\frac{2529092855837558}{2690148205992001}$ | $\frac{37657795243316428}{38019573842698655}$ | AR1_REPLAY |
| | open | $\frac{37657795243316428}{38019573842698655}$ | $\frac{1}{1}$ | KNN, frozen_F, frozen_S |
| | point | $\frac{0}{1}$ | | QR |
| | point | $\frac{8631914270543611}{12443891775419019}$ | | QR, frozen_K |
| | point | $\frac{2529092855837558}{2690148205992001}$ | | AR1_REPLAY, frozen_K |
| | point | $\frac{37657795243316428}{38019573842698655}$ | | AR1_REPLAY, KNN, frozen_F, frozen_S |
| | point | $\frac{1}{1}$ | | KNN, frozen_F, frozen_S |
| Bills sparse H24/W260 | open | $\frac{0}{1}$ | $\frac{19956537365855950}{27652346249496953}$ | frozen_K |
| | open | $\frac{19956537365855950}{27652346249496953}$ | $\frac{1}{1}$ | AR1_REPLAY |
| | point | $\frac{0}{1}$ | | frozen_K |
| | point | $\frac{19956537365855950}{27652346249496953}$ | | AR1_REPLAY, frozen_K |
| | point | $\frac{1}{1}$ | | AR1_REPLAY |

## E3 Numerical implementation and inference conventions

**Completed labels and update clocks.** At a scheduled refit $\kappa$, the latest mature peak response origin is $\kappa - H$. A history of $W$ mature peak responses therefore uses $\{\kappa - H - W +$

$1, \ldots, \kappa - H\}$, with every target ending by that refit. Training transforms, scales, coefficients and historical pools are held until the next scheduled refit. The current observed position may still change at each forecast issue. Replacing that issue anchor by the refit anchor changes the forecasting procedure.

**Regression and local quantiles.** The regression objective uses a training-normalized headroom response, an unpenalized intercept and a ridge penalty on slopes. Projection is applied when issuing the report and evaluating its validation score, not inside the fitted training objective. Local methods use training-only standardized predictors, fixed neighbour menus and duplicate-preserving order statistics. A selected whole-pool local quantile is an exact historical identity, not a failed-fit fallback. Scale-adjusted neighbours change responses as well as retrieval.

**Numerical eligibility.** The accepted solver records primal and dual residuals, objectives, feasibility and their gap. Ill-conditioned, nonfinite, unsupported-rank and failed candidate states retain their original statuses under each named selection contract. A mandatory-candidate failure blocks a procedure when its contract requires every candidate. The original full regression has that requirement; the four-feature selector may choose among completed eligible candidates. A failed selected evaluation never receives a replacement penalty or a shortened score. Changing the ridge menu changes the procedure even when feature definitions stay fixed. A numerical certificate does not establish financial accuracy, economic relevance, inferential validity or production continuity. No production continuity or fallback policy is evaluated here.

**Working-reference inference.** Saved unconditional loss differences use the recorded Bartlett-HAC convention; HLN and doubled-bandwidth sensitivities are labelled separately. Separate families and later reporting partitions do not erase the wider selection history. Coverage references distinguish full-grid Kupiec from the transition-conditional Christoffersen joint likelihood, which conditions on the initial hit. Small sparse grids and boundary likelihoods retain their original status. Neither a nonrejection nor a small working-reference p value proves conditional coverage or supplies selection-adjusted confirmation.

**Common-pool construction distances.** On one nonempty historical episode set, compare the historical headroom, translated-path and historical-peak reports using the same available order statistic. Each episode peak includes its starting observation. A translated path changes each level by the same anchor difference, so its absolute peak changes by at most that difference. Flooring is nonexpansive and commutes with the fixed order statistic. The headroom construction has the corresponding bound in the difference of anchor magnitudes. Thus

$$|C^{\text{headroom}} - C^{\text{peak}}| \le \max_i \big||N_i| - B\big|, \qquad |C^{\text{translated}} - C^{\text{peak}}| \le \max_i |N_i - x|.$$

For a proof of the order-statistic step, paired coordinatewise differences bounded by a radius imply that each empirical quantile lies below the other plus that radius; reverse the roles for the second inequality. These deterministic same-pool bounds do not order financial accuracy or coverage. Changing historical rows on one side invalidates the common-pool argument.

**Complete implementation evidence.** Exact menus, numerical tolerances, failures and diagnostic tables remain in EXTENDED RESULTS and the dated source archives. The separate study receipts identify the executed additions. This document builder imports no forecasting code and only formats unchanged public results.